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Inertial Relativity: Demonstrating the Nature of, and Connection between, Special and General Relativity

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ABSTRACT: Inertial Relativity and its key formula the Space-Time Equivalence (STE)[1], state that there is a universal mathematical relationship between time and the spatial distribution of matter: The ratio of clock rates between any two physical systems about ANY respective axes is universally equal to the fifth root of the inverse ratio of their moments of inertia about the same axes. In our prior paper [2], we demonstrated this universality by arriving at General Relativity calculations with exact precision utilizing only principles of relativistic inertia. This paper closes the loop by exploring Inertial Relativity's mathematical relationship with General and Special Relativity. In the context of moment of inertia about an axis, we demonstrate that the STE is mathematically equivalent to Schwarzschild gravitational time dilation. In the context of linear inertia (mass), the STE states an inverse third root relationship with inertia (mass), which we demonstrate is mathematically equivalent to Lorentz time dilation. Via this exploration, we show that Special and General Relativity emerge as edge cases of the same Underlying Phenomenon, Inertial Relativity. Special Relativity at the extremes of velocity and General Relativity at the extremes of mass, thus unifying the two under a single universal relativistic framework.

1 Introduction

This paper presents a mathematical derivation connecting The space-time equivalence (STE)[1], General Relativity[3], and Special Relativity[4]. The work in this paper builds on the Space-time Equivalence (STE). We then build upon DeGerlia's immediate prior work introducing the concept of inertial density and predicting time dilation calculations with exact precision, introduced in a prior paper [2], which established that the Schwarzschild condition reduces to a constant threshold of mass over radius — the DeGerlia Threshold: $D_{crit} = 6.73295 \times 10^{26}$ kg/m — about the selected axis. And from these demonstrations the mathematical equivalence between the STE and General and Special Relativity is established, unifying the two theories under a single framework: Inertial Relativity.

2 Inertial Relativity

Inertial Relativity is a principle introduced by the author in a prior paper "the Universe of Light" [1]. In that work, we hypothesized that Inertial Relativity is relativity from which general and special relativity emerge. Inertia is relativistic. Meaning that without a comparator, any measurement of it would have no meaning. For example, something with a moment of inertia 1×10^{20} kg/m^2 might be perceived as difficult to rotate. But not more difficult than something with 1×10^{30} kg/m^2. However, the concept of relativity also says that how something is observed is influenced by the frame of observation. Consider for a moment, the following statement that exemplifies our nuanced understanding of the concept of relative scale:

> "When we were kids, adults looked big. Now that we're adults, kids seem tiny."

This illustrates that we understand what spatial scale means, even though we have not necessarily defined it mathematically. We know what relative scale means; we know what it's like to be bigger than something or to be smaller than something. We understand the experience of growing or watching something grow; we know what it is like to change scale relative to something else. Further, we know that any amount of mass can accompany any amount of spatial distribution from zero to infinity. As such, there is little meaning to evaluating mass alone.

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2.1 Rotational Inertia

Rotational inertia is called moment of inertia. Moment of inertia for a system is distinct about every axis in the system, and for the axis of rotation that moment of inertia dictates how force translates into a pace of motion. In fact, the axis with the least moment of inertia will be the axis of rotation. It dictates what delta in motion results from an applied torque. Or about the axis of observation? Moment of inertia is defined as:

$$I = \sum_i m_i r_i^2 \tag{1}$$

where r_i is the perpendicular distance from each mass element m_i to the chosen axis. Just as mass governs how a body responds to linear force ($F = ma$), moment of inertia governs how it responds to torque ($\tau = I\alpha$).

2.2 Linear Inertia

Linear inertia is mass:

$$m = \sum_i m_i \tag{2}$$

where m_i are the constituent mass elements.

A body, in this context, is any system of mass you choose to define. Linear inertia is the resistance of a body to changes in its linear velocity, quantified entirely by its mass ($F = ma$). It is dimensional in the sense that it acts along a specific vector — the direction of the applied force — but unlike moment of inertia, it is isotropic: mass offers the same resistance to acceleration regardless of which direction that vector points.

2.3 Isometric Scaling

Isometric scaling defines k as the linear scale factor:

$$L' = kL \tag{3}$$

where L is the characteristic length of the system and k is the linear scale factor. Isometric scaling of a system by a factor of k results in the following changes to mass, radius, and moment of inertia:

$$M' = k^3 M \tag{4}$$

where mass is M and k is the linear scale factor.

$$R' = kR \tag{5}$$

$$I' = k^5 I \tag{6}$$

where I is the moment of inertia about a selected axis; k is the linear scale factor [5].

$$I'_{lin} = M' = k^3 M \tag{7}$$

where I_{lin} is linear inertia (mass); k is the linear scale factor.

Because every change in length necessarily changes the system's mass, the first two formulas hold only for strict isometric scaling; the third abstracts mass and radius as moment of inertia. If two systems that share moment of inertia about any axis, then an identical torque applied to each system about that same axis will produce identical changes in motion, regardless of the system geometry that arrived at that moment of inertia; absolutely independent of mass and radius individually. The moment of inertia alone dictates how a system interacts with time about a particular axis. From this, we can conclude the following formula is universal and is not bound in any way to isometric scaling. We demonstrate the validity of the STE herein by demonstrating that general and special relativity emerge from this principle.

2.4 Universality

The extension from isometrically scaled systems to all systems follows from Premise 2. Because I fully determines rotational response to applied torque, two systems with the same I are dynamically indistinguishable regardless of geometry. A thin shell and a solid sphere with the same I exhibit the same rotational physics. Since clock rate is a dynamical observable (Premise 3), it cannot depend on geometric details that I has already absorbed.

The fifth root relationship therefore holds universally: for any two systems, the ratio of their clock rates is the fifth root of the inverse ratio of their moments of inertia. Premise 1 establishes the exponent. Premise 2 establishes universality. Premise 3 establishes that clock rates are within the domain of dynamical scaling.

2.5 The Space-Time Equivalence (STE)

The STE is the core universal equivalence stated by Inertial Relativity [1]. Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{8}$$

where k is the linear scale factor derived from inertia. In the general case:

$$k = \left(\frac{I_1}{I_2}\right)^{1/5} \tag{9}$$

where I_1 and I_2 are the moments of inertia of any two systems about any respective axes.

For linear (non-rotational) systems, the R^2 component of moment of inertia is absent, so inertia reduces to mass:

$$k = \left(\frac{M_2}{M_1}\right)^{1/3} \tag{10}$$

These equations reveal the following about time:

- It is relativistic, it only has meaning when compared, how it appears depends on the frame of observation
- It is emergent of inertia
- It is dimensional in three spatial dimensions
- It is multimodal: both linear and rotational
- It is dynamic for every system or subsystem
- It is infinitely precise
- It applies to all systems, not only uniform or spherically symmetric ones

- It is infinite
- It is universally and precisely determined by the spatial distribution of matter in the system, as captured by inertia, about and along every axis, whether or not in motion or being observed
- General relativity reflects relative rotational inertia or moment of inertia (mr^2)
- Special relativity reflects linear inertial scale (mass)

At any given time, any system is always undergoing some degree of rotation and some amount of linear motion, however small that is. It also demonstrates that the basic gravitational time dilation model accommodates for distorted black holes such as the Kerr singularity. The STE states that the relative pace of time for any system about any axis is determined by its moment of inertia about that same axis. The more inertia a system has about that axis, the slower its clock runs relative to a system with less inertia. This is consistent with the notion that time is a measure of change, and inertia resists change. Therefore, the more inertia a system has, the less it changes over a given period of time, and thus the slower its system clock runs relative to a system with less inertia. Note that this law applies to any system whose mass is distributed over some region of space — requiring, at minimum, both mass and spatial extent. One may define the system as the core of a planet, a disc-shaped galaxy, a cosmic filament, or a spherical sample of interstellar space.

2.6 Inertial Density

Inertial density P is the moment of inertia of a system divided by its volume [2]:

$$P = I/v \tag{11}$$

Since linear inertia is mass, this reduces directly to ordinary density $\rho = m/v$ in the linear case — inertial density is the rotational generalization of a quantity already familiar from linear mechanics.

For a uniform sphere, substituting $I = \frac{2}{5}mr^2$ and $v = \frac{4}{3}\pi r^3$, this reduces to:

$$P = \frac{m}{r} \times \frac{3}{10\pi} \tag{12}$$

Just as ordinary density ($\rho = m/v$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in [2]: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.* Because P is defined relative to a selected axis, it can be evaluated per axis directly rather than requiring a full tensor treatment when the quantity varies by direction.

DeGerlia Compactness $D = m/r$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system about the selected axis.

Setting $r = r_s$ (the Schwarzschild condition) and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia Compactness at which a system undergoes gravitational collapse:

$$D_{crit} = \frac{c^2}{2G} = 6.73295 \times 10^{26}\ \text{kg/m} \tag{13}$$

The ratio of a system's DeGerlia Compactness to this threshold gives the time dilation directly:

$$d\tau/dt = \sqrt{1 - D/D_{crit}} \tag{14}$$

2.7 STE Example

If one takes a uniform sphere of 1 m in radius and 1 kg of mass and scales it up by a factor of two, the resulting system is 8 kg of mass with a radius of 2 m. Moment of inertia of the initial system is $\frac{2}{5} \cdot 1\,\text{kg} \cdot (1\,\text{m})^2 = 0.4\,\text{kg}\,\text{m}^2$. That larger scale system has a mass that is the cube of the scale factor multiplied by the original mass $2^3 \cdot 1\,\text{kg} = 8\,\text{kg}$, and would have a moment of inertia that is the fifth power of the scale factor times the original moment of inertia. The inertial scale factor is $k^5 = 2^5 = 32$. Conservation of momentum gives the best representation of how changes in moment of inertia affect motion. The moment of inertia for the scaled system calculates to $\frac{2}{5}(8\,\text{kg})(2\,\text{m})^2 = \frac{2}{5} \cdot 8\,\text{kg} \cdot 4\,\text{m}^2 = 0.4(32)$, resulting in the predicted factor of two to the fifth power, or 32. If our sphere were rotating at one meter per second initially, the scaled system would now be moving half of that (0.5 m/s) resulting from the $2^5 = 32\times$ increased moment of inertia.

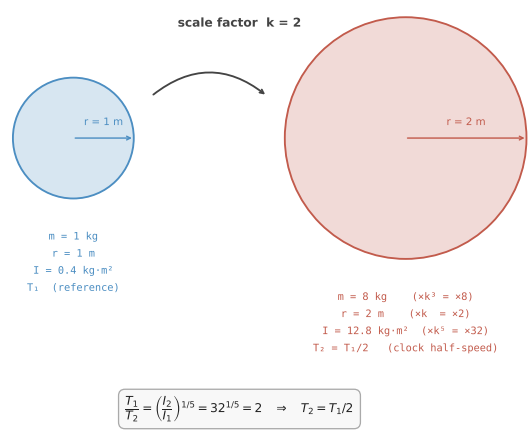

Fig. 1 Isometric scaling of a uniform sphere by factor $k = 2$. Mass scales as k^3, moment of inertia as k^5, and the clock rate ratio follows directly from the STE.

3 Expressions of k

The parameter k is always an inertial ratio — a ratio of inertial quantities. What varies is which components of inertia are already known or fixed. The following expressions each derive k from different known quantities.

3.1 Raw Time Dilation

$$\text{RTD} = \sqrt{k}, \qquad \text{RTD}^2 = k \tag{15}$$

3.2 STE (Full Inertia)

$$k = \left(\frac{I_1}{I_2}\right)^{1/5} \tag{16}$$

For uniform spheres:

$$k = \left(\frac{\frac{2}{5}m_2 r_2^2}{\frac{2}{5}m_1 r_1^2}\right)^{1/5} \tag{17}$$

3.3 Static Density (Isometric)

$$k = \left(\frac{M_2}{M_1}\right)^{1/3} \tag{18}$$

3.4 Static Mass (Gravitational)

$$k = \frac{R_S}{R} \tag{19}$$

$$R' = kR \tag{20}$$

$$R_S = kR \tag{21}$$

The Schwarzschild radius is $r_s = 2GM/c^2$. The ratio r_s/r factors as:

$$r_s/r = (2G/c^2) \times (M/R) \tag{22}$$

The reciprocal of $2G/c^2$ is $c^2/2G$ — the DeGerlia Threshold. Therefore:

$$k = r_s/r = (M/R)/(c^2/2G) \tag{23}$$

This is an algebraic identity. The Schwarzschild ratio and the inertial density threshold ratio are the same expression.

3.5 Inertial Density

$$k = \frac{I_2/V_2}{I_1/V_1} \tag{24}$$

3.6 Mass–Radius Ratio

$$k = \frac{M_2/R_2}{M_1/R_1} \tag{25}$$

$$k = \frac{M_2/R_2}{6.73295 \times 10^{26}} \tag{26}$$

3.7 Special Relativity (Static Radius)

$$k = \frac{v^2}{c^2} \tag{27}$$

In isometric form:

$$k = \left(\frac{m_2}{m_1}\right)^{1/3} \tag{28}$$

In special relativity, motion is linear rather than rotational. Linear inertia is mass alone (no R^2 component). Mass has three dimensions of length-equivalent content (mass = density × volume, volume = length^3). Extracting linear scale therefore requires a cube root, not a fifth root.

3.8 Unification of GR and SR

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{29}$$

and k is always the linear scale factor derived from inertia. The different expressions of k presented above are not different formulas — they are the same inertial ratio with different components already resolved. When all components of inertia are present, $k = (I_2/I_1)^{1/5}$. When density is fixed, k reduces to $(M_2/M_1)^{1/3}$. When radius is fixed, $k = R_S/R$. When radius drops out entirely (linear motion), inertia reduces to mass and $k = (m_2/m_1)^{1/3}$.

General relativity addresses the rotational case, where inertia is moment of inertia ($I = mr^2$) with five dimensions of inertial content. Special relativity addresses the linear case, where inertia is mass alone with three dimensions of inertial content. The distinction between the two theories maps directly onto the distinction between rotational and linear inertia. They differ by the R^2 component of moment of inertia and by nothing else.

4 Analysis and Conclusion

All formulations of relativistic time dilation derive the same linear scale factor k:

Inertial Scaling Law:

$$T_1/T_2 = (I_2/I_1)^{(1/5)} \tag{30}$$

gives the time dilation factor directly as a ratio of inertial properties, derived from isometric scaling, the completeness of moment of inertia, and the identification of clock rates as dynamical observables governed by inertial content.

Inertial Density Ratio:

$$k = (M/R)/(c^2/2G) \tag{31}$$

gives the fraction of the universal Schwarzschild threshold consumed by the system, and is algebraically identical to r_s/r.

Schwarzschild Formula [6]:

$$d\tau/dt = \sqrt{1-k} \tag{32}$$

gives the time dilation factor through Einstein's quadratic geometric formalism [3].

Raw Time Dilation:

$$\text{RTD} = \sqrt{k} \tag{33}$$

gives the direct inertial fraction without the Schwarzschild normalization.

In the linear case (special relativity), the R^2 component of moment of inertia is absent, so inertia reduces to mass and the fifth root becomes a cube root. General and special relativity are the same inertial scaling law applied to two types of inertia.

The algebraic equivalences hold as mathematical identities. They produce exact agreement with both Schwarzschild and Lorentz time dilation, with no free parameters. The equivalence with Schwarzschild constitutes equivalence with the field equation solution by transitivity, and is evaluated against the same

standard applied to the Schwarzschild solution itself: exact agreement for the configuration it addresses.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste`.

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